SAE & Causal Analysis of Malaria Interventions in West Africa

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Background and Problem

Malaria remains a public health challenge in West Africa, yet accurate data at the district level is often scarce. While Demographic and Health Surveys provide essential data on prevalence and intervention coverage, they are designed for national-level precision[1]. When disaggregated to the district, these direct estimates rely on small sample sizes, resulting in noisy estimates with high variability[2].

To overcome this data scarcity, we employ the Fay-Herriot model for small area Estimation. This approach borrows strength from auxiliary data, such as census demographics and climate variables to generate smoothed, robust estimates of malaria burden and intervention coverage for every district. Beyond estimation, we leverage these values to investigate the causal effect of interventions, adjusting for confounding variables to model counterfactual scenarios and project how different strategies could reduce the future malaria burden[3].

Objectives of Project

To combine survey, census and climate data to make Small Area level estimates of malaria incidence, ACT uptake and ITN coverage in select west African countries using the Fay-Herriot model.

To use the estimate obtained from SAE to estimate the casual effects of malaria interventions (ACT, ITN, Medical treatment for fever) on small area level malaria burden while adjusting for confounding variables

To use the causal estimate to carry out projections based on different counter factual scenarios (e.g., if 80% of the people were sleeping under nets)

Data Sources and Datasets

Primary Survey Data (DHS Program)

Environmental Predictors (Malaria Atlas Project and DHS): High-resolution raster data including precipitation, mean temperature, altitude, and proximity to water bodies.

Socio-Demographic Factors (HDX): Census-derived population density, poverty mapping, and urban/rural classifications.

Spatial Structure (DHS & WHO):Household geolocations boundaries from the DHS and official country and Administrative Level 2 (District) boundaries from the WHO Health Geographics hub.

Descriptive Statistics of Datasets

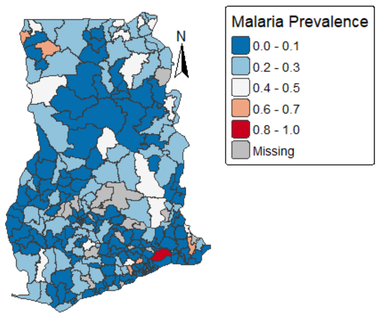

Figure 1: SE vs Sample Size (Malaria Prevelance)

While the map suggests low prevalence in many districts, the Reliability Plot below confirms these are artifacts of high error (CV > 30%) due to small sample sizes, necessitating SAE to 'borrow strength' and correct the estimates.

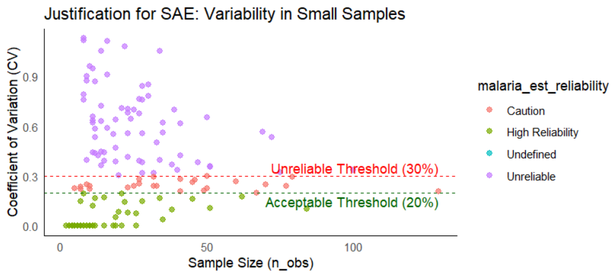

Figure 2: SE vs Sample Size (Malaria Prevelance)

The plot above shows why direct estimates fail. As sample size shrinks, the CV goes up, pushing most district estimates above the 30% reliability threshold. In these high noise districts, the SE is very large that the estimates become usable for policy making.

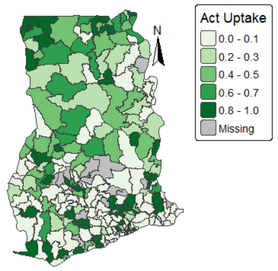

Figure 3: SE vs Sample Size (Malaria Prevelance)

The same is true for the uptake of Artemisinin-based combination therapies. The map above suggests districts with greener shades have high uptake of ACT thus low prevalence of malaria. The plot below is trying to then give the reality of the direct estimates used in coming up with map above.

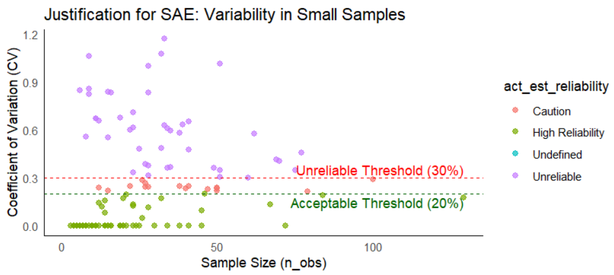

Figure 4: SE vs Sample Size (Malaria Prevelance)

Next Project Steps

Small Area Estimation
Smooth noisy direct estimates using a Poisson regression variant of the Fay-Herriot model in R. Compare model specifications for independent versus spatial random effects to account for neighboring district similarities.

Causal Inference
Use the smoothed small area estimates to determine the causal effect of interventions on malaria burden, adjusting for confounding variables to isolate the true impact.

Counter factual Projections
Perform analysis to predict malaria burden under hypothetical scenarios (e.g., increasing ITN coverage to 80%) and develop an interactive Shiny App to visualize current versus projected outcomes.

GitHub

The code and datasets for this project can be viewed at our GitHub repository here: https://github.com/SAHagen/HDS5106

References

1. Corsi, D. J., Neuman, M., Finlay, J. E., & Subramanian, S.. (2012). Demographic and health surveys: a profile. International Journal of Epidemiology, 41(6), 1602–1613. https://doi.org/10.1093/ije/dys184

2. Rao, Jnk, and Isabel Molina. Small Area Estimation /. Second. Hoboken, New Jersey : Wiley, 2015. Print.

3. Weiss, D. J., et al. (2019). "Mapping the global prevalence, incidence, and mortality of Plasmodium falciparum, 2000–17: a spatial and temporal modelling study." The Lancet 394(10195): 322–331.